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Paleomagnetism of Early Cambrian Itabaiana mafic dikes (NE Brazil) and the final assembly of Gondwana

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Abstract

Paleomagnetic analysis on 15 early Cambrian mafic dikes from Itabaiana (Paraíba State) yielded a southern (northwestern) direction with steep upward (downward) inclination (D_m=167.5°, I_m=−63.7°, α_{95}=7.3°). AF and Thermal demagnetization, thermomagnetic curves, and hysteresis results suggest that this component is dominantly carried by fine-grained SD magnetite. The high stability of this component and positive baked contact tests on three dikes indicate it represents a primary thermoremanent magnetization. Ar–Ar analysis on whole-rock samples from two sites provides a strong constraint on the age of the Itabaiana paleomagnetic pole (134.6° E, 34.9° S; A_{95}=7.3, K=28) defined by plateau ages of 525±5 and 526±4 Ma. This pole completely satisfies six out of the seven quality criteria proposed by Van der Voo [R. Van der Voo, The reliability of paleomagnetic data, Tectonophysics 184 (1990) 1–9.] and permits a tight constraint on the Early Cambrian sector of the Gondwana apparent polar wander path. Paleogeographic reconstructions consistent with the available paleomagnetic and geological record show that Gondwana was sutured along three major orogenies, the Mozambique (Brasilano/Pan-African) Orogeny (800–650 Ma), the Kuunga Orogeny (570–530 Ma) and the Pampean–Araguaia Orogeny (540–520 Ma). We suggest that after rifting away from Laurentia at the end of the Neoproterozoic, opening the Iapetus ocean, the Amazonian craton and minor adjoining blocks, such as Rio Apa and Pampia, collided with the proto-Gondwana by Cambrian times at ca. 530–520 Ma. Unless for small adjustments, Gondwana was completely formed by 525 Ma whose paleogeography is defined by the Itabaiana pole.
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Keywords: Gondwana; Cambrian; Neoproterozoic; paleomagnetism; paleogeography

1. Introduction

The building of the supercontinent Gondwana (Fig. 1a) is classically viewed as the assemblage of its two halves – the east and west Gondwana – along the 800–650 Ma Mozambique belt [2]. In such a model the eastern and western halves of Gondwana would form two large, separate blocks prior to its final collision.

For the last years, however, the idea of a united East Gondwana has been challenged by new isotopic and paleomagnetic data [3,4] which show that their constituting pieces were assembled along diachronic

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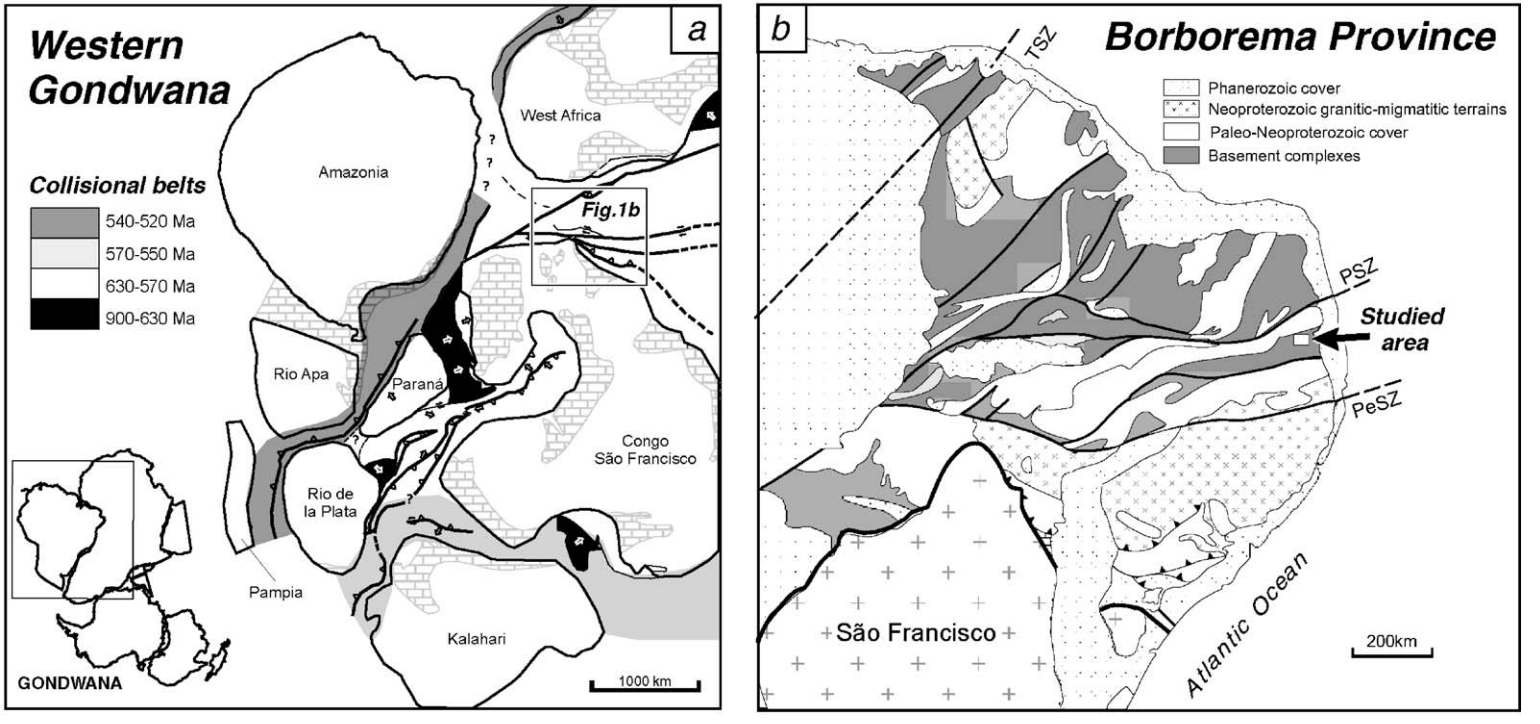


Fig. 1. (a) Western Gondwana at the end of Neoproterozoic times with indication of main orogenic belts and respective collisional ages (modified from [7]). (b) Borborema Province with indication of main shear zones (PSZ: Patos, PeSZ: Pernambuco, TSZ: Transbrasiliano). The Transversal Domain is comprised between PSZ and PeSZ.

collisions, spanning in time the whole Neoproterozoic and entering into the Cambrian. This long span of time encompasses at least three major episodes of accretion, including the formation of the Arabian–Nubian arcs at 820–750 Ma, the East Africa (or Mozambique) Orogen at 650–620 Ma and the Kuunga Orogen at 550–530 Ma [4].

As for the western half of Gondwana (Fig. 1a), the idea of a diachronic build-up starts to emerge from recent, more precise geochronological data. A recent compilation of geochronological data for South America [5] suggests a long assembly process, taking place from 880 up to 500 Ma ago. This is in clear conflict with the concept of a synchronous assembly by around 600 Ma, in a unique orogenic event, the so-called Brasiliano Orogeny [6]. Instead, at least two major assembly events are recognized during this long age interval [7–12]. The first encompasses a series of collisions at the western and northern margins of the Congo–São Francisco craton resulting in closure of Neoproterozoic oceans in the 880 to 600 Ma interval [8,9,11]. After that, tectonic activity around the Congo–São Francisco craton relates mostly to large-scale strike-slip shear zones, coeval with granitoid magmatism dated from 630 up to 570 Ma. The Borborema Province (Fig. 1b), in Northeast Brazil, and its African counterparts in Nigeria and Hoggar typify the tectonic evolution of the collisional belts encircling the São Francisco craton [13–15]. Nevertheless, the margins of Kalahari, Rio de la Plata and Amazon cratons, as well as the southern border of the São Francisco craton, record much younger collision events, ageing up to Cambrian times (Fig. 1a). In spite of their similar ages all over South America, these Cambrian events are usually viewed as of local extent. They have been named Buzius Orogeny in southeastern Brazil [10], and Pampean Orogeny in western Argentina [16].

Paleomagnetism provides key information to constrain such a complex assembly history. However, the scarcity of paleomagnetic data for late Neoproterozoic and Cambrian rocks in the West Gondwana has prevented testing these geodynamic scenarios (for a review, see [17]). Here we present a high-quality paleomagnetic pole obtained on the Early Cambrian mafic dikes of Itabaiana, Paraíba State, NE Brazil. These results are then compared with data from Gondwana, Laurentia and with poles recently obtained for the Amazon craton [18]. This new dataset allowed us to generate paleogeographic models for the end of Neoproterozoic and Cambrian times, tracing the final assembly of the Gondwana supercontinent.

2. Geological setting and sampling

The Borborema Province in Northeast Brazil and its counterpart in Africa, is limited to the north by the West Africa craton and to the south by the Congo–São Francisco craton (Fig. 1). They form a huge branch of Neoproterozoic fold belts comprising basement inliers of Archean to Paleoproterozoic age, surrounded by Meso- to Neoproterozoic sedimentary basins, which were intensely reworked between 730 and 550 Ma ago [7,19]. Cooling and terrain exhumation following this event have been completed before 520 Ma [20].

In the eastern Borborema Province, two E-trending continental-scale shear zones bound four strips of geologically contrasted terrains, grouped into the Transversal Domain (Fig. 1b). The Itabaiana dikes intrude the Paleoproterozoic peraluminous orthogneisses of the Alto Moxotó Terrain (U–Pb zircon ages between 2.01 and 2.23 Ga; [21,22]), a basement slice sandwiched between two Neoproterozoic belts within the Transversal Domain. The dike swarm crops out along the Paraíba River, close to Itabaiana city, Paraiba State, Brazil (Fig. 2). Here, it is preliminarily divided into two main groups according to their petrography and strike; geochemical studies are under progress. Most dikes show NW–SE strikes and thickness ranging from 15 cm up to 1 m. They are composed of microphaneritic olivine–basalt and dolerites. Some dikes (site #7) show N–S strike and thickness around 5–10 cm. They are composed of vitreous to aphanitic basalt, and cut the former ones, sometimes incorporating olivine xenocrysts. A shallow intrusion level for the two sets of dikes

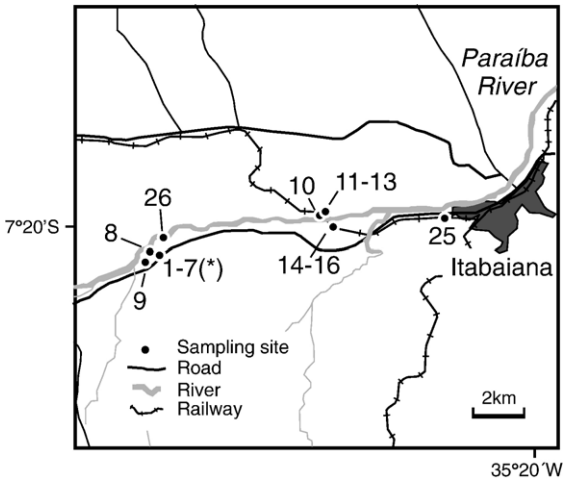


Fig. 2. Location of the Itabaiana dike swarm, with indication of paleomagnetic and geochronologic sampling sites. (*) Site 7 corresponds to a N-trending dike. Sites 1 and 3 were also sampled for Ar–Ar dating (samples 490-02 and 490-03, respectively).

is indicated by the presence of chilled margins at their sharp, straight contacts with the host gneisses. Fifteen dikes were sampled for paleomagnetic purposes at three different sectors (Fig. 2). The sampling included the center and the borders of the dikes. Samples from the host gneisses were collected across the margins of three dikes with thickness varying from 20 to 35 cm (sites #1, #8 and #26) in order to perform a baked contact test. In addition, samples from the host rocks were collected in a control site (site #9) far from any dike. At two sites (sites #1 and #3) hand samples of fine-grained basalt were also collected for Ar–Ar dating (Fig. 2).

3. Paleomagnetism

Paleomagnetic samples were oriented in the field by sun and magnetic compasses. In laboratory, samples were cut into 22×25 mm cylindrical specimens and submitted to thermal or alternating field (AF) demagnetization along 16 to 24 progressive steps. Paleomagnetic analyses were performed in the Paleomagnetic Lab of the Universidade de São Paulo. Magnetic remanence was measured using a Molspin-fluxgate spinner magnetometer or a 2G-cryogenic magnetometer. Alternating field (AF) demagnetization was done by a Molspin tumbler–demagnetizer or by an automated three-axis AF-demagnetizer coupled with the cryogenic magnetometer. Thermal demagnetizations were performed with a Magnetic Measurements oven (MMTD60; peak temperatures within ±2 °C, heating time of 1 h). These instruments are housed into a magnetically shielded room with ambient field <1000 nT. Magnetic components for each specimen were identified in orthogonal plots, and calculated using least-squares fits [23]. Fisherian statistics was used to calculate vector mean directions and paleomagnetic poles. A total of 161 specimens of mafic dikes were analyzed. Natural remanent magnetizations (NRM) in these specimens ranged from 10^{-1} to 15 A/m. In addition, 28 specimens from the host gneisses were analyzed, with NRM values between 10^{-4} and 0.3 A/m.

3.1. Mafic dikes

Thermal and AF treatments isolated a single, very stable characteristic component in all samples from mafic dikes after removing a weak, randomly oriented viscous component (Figs. 3 and 4). A steep, negative remanence vector, with declinations varying from south to southeast was disclosed for samples from all sites but one site of olivine–basalt (site #25). Samples from site #25 have shown a steep, positive vector directed to the northwest (Fig. 3b). Sample and site–mean directions are shown in Fig. 5. After inversion, the directions obtained in site #25 lie close to the south to southeast dipping directions obtained in the other sites (Fig. 5a), but they fail both site and sample-based reversal tests [24]. We note however that the virtual geomagnetic poles (VGPs) derived from samples of site #25 do not match younger poles of the Gondwana database [25]. Moreover, they do not match poles from South America after the opening of the South Atlantic Ocean [26]. In addition, they fall within 30° from the mean pole obtained from the other sites. Mean direction and VGP calculated for each site and the Itabaiana mean direction and paleomagnetic pole, including (or not) the results of site #25 after inversion, are given in Table 1.

The complete demagnetization for most of the samples was attained at relatively high peak fields between 50 and 120 mT (Fig. 3a–d). The unblocking temperatures are distributed within a narrow range between 540° and 560 °C, and thermal demagnetization curves show a 'hard-shoulder' shape (Figs. 3e and 4g) suggesting that fine-grained single-domain (SD) (titano) magnetite is the main remanence carrier in these rocks. At some samples, a low-coercivity component was also identified, which is probably carried by coarse-grained multidomain (MD) grains. The presence of dominantly fine-grained SD magnetite, but also coarse-grained MD magnetite at some samples, is further confirmed by low-temperature thermomagnetic curves, hysteresis ratios and other magnetic tests. In thermomagnetic curves (Fig. 6a) samples from mafic dikes systematically show both the Verwey transition at around −150 °C and a strong drop in magnetic susceptibility between 560–580 °C. Most curves present a non-reversible behavior, with an increase in magnetic susceptibility after heating up to 700 °C. In the Day diagram (Fig. 6c), hysteresis ratios fall very close to the SD end-member into the SD–MD mixing curve of Dunlop [27]. In addition, the strong asymmetry in isothermal remanence (IRM) acquisition and demagnetization curves [28] suggests a high degree of magnetic interaction between fine-grained magnetic grains in these rocks (Fig. 6d).

3.2. Host rocks and baked contact test

The host gneisses sampled far from mafic dikes have usually shown only a weak, random magnetic component (Fig. 5b). Most samples are completely demagnetized at AF peak fields of 20 to 30 mT (Fig. 3f). Under thermal treatment these samples were gradually demagnetized up to 540–560 °C (Fig. 4i), showing the long demagnetization tails that characterize multidomain

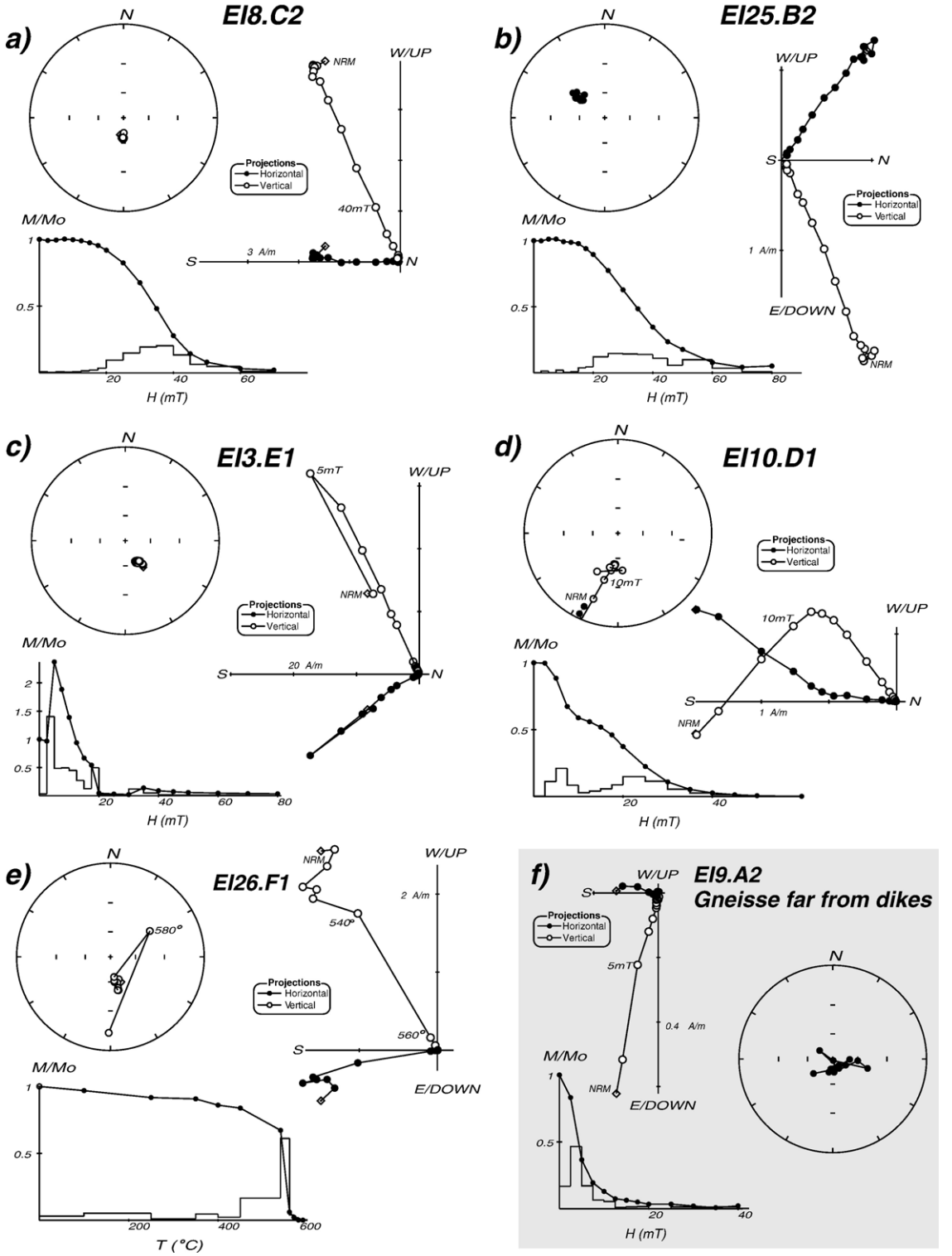


Fig. 3. Typical demagnetization trajectories for mafic dikes (a–e) and host gneisses (f: gray rectangle). Orthogonal and stereographic projections, magnetization intensity decay curves, and coercivity and unblocking temperature spectra are represented for each sample. Open (full) symbols correspond to negative (positive) inclinations in the stereographic projections.

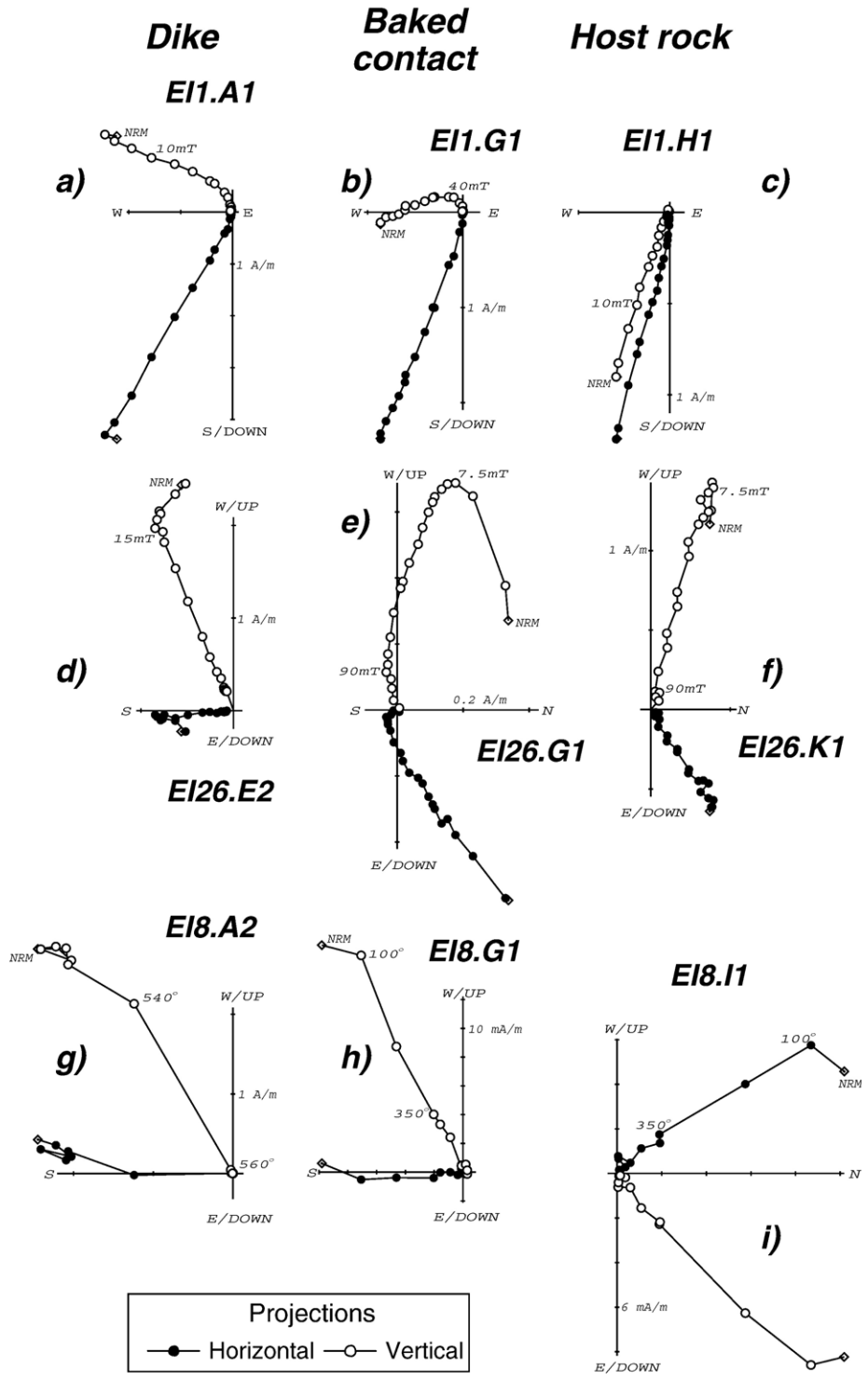


Fig. 4. Baked contact test for sites 1, 8 and 26. Note the different directions isolated from Itabaiana dikes (a,d,g) and the host rocks far from the dikes (c,f,i) at each site. The gneisses collected at the contact show these two directions (b,e) or only the direction of the dike (h).

magnetite [29]. Despite the poor quality of the magnetic record of most host rocks sampled far from any dike, the host rocks sampled at the contact of the intrusions have systematically shown the characteristic component of the dike, hence providing a positive baked contact test (Fig. 4). After thermal treatment, the host rock sample

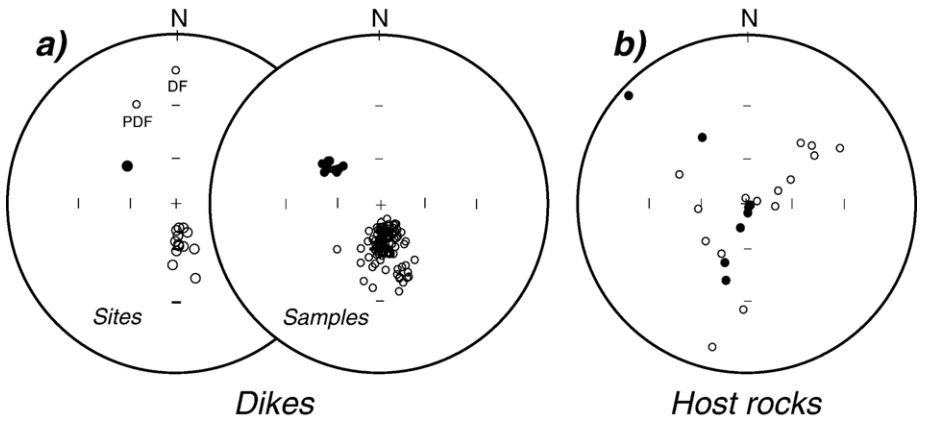


Fig. 5. Sample and site–mean directions for (a) dikes and (b) host rocks. Open (full) symbols correspond to negative (positive) inclinations. PDF: present day geomagnetic field; DF: dipolar field.

EI8.G1, collected 1 cm away from the 13 cm thick EI8 dike, yielded a single component which is identical to that disclosed for the dike itself (Fig. 4g–h). This direction contrasts with that obtained for the host rock 20 cm away from the dike (Fig. 4i). At sites #1 and #26, the AF treatment also isolates the same characteristic component for both the dike and the host rock at the contact (Fig. 4a–f). The thermal effect of the dikes is also indicated by changes in the host-rock's magnetic mineralogy in the vicinity of the intrusions. When approaching the dikes, both the intensity of natural remanent magnetization (NRM) and the medium destructive field ($H_{1/2}$) strongly increase (Fig. 7) suggesting that new, highly coercive magnetic carriers are formed due to the thermal influence of the dike. Accordingly, most thermomagnetic curves for host rock samples show non-reversible behavior, with formation of new magnetic phases after heating (Fig. 6b). This newly formed high coercivity fraction is likely the carrier of the more stable remanence in the baked contact rocks which mimics the characteristic component of the dikes.

Table 1
Paleomagnetic results

Site#	N	Site mean direction				VGP	
		Dec. (°)	Inc (°)	α_{95}	k	Plong. (°E)	Plat. (°N)
1	10	182.9	−50.3	9.0	30	149.2	−50.9
2	9	175.5	−68.7	5.1	99	141.3	−30.2
3	6	157.4	−57.1	7.8	77	121.3	−40.0
4	11	161.6	−73.3	3.4	180	134.5	−21.8
5	6	174.2	−72.7	2.3	862	141.2	−24.3
5A	9	174.5	−72.6	2.7	367	141.5	−24.4
6	12	168.3	−74.1	2.8	247	138.3	−21.4
6A	5	157.5	−68.3	7.1	118	128.6	−27.9
7	9	176.0	−65.9	3.6	203	141.7	−34.8
8	11	177.1	−63.1	3.3	195	142.0	−37.6
10	10	179.1	−58.9	3.1	249	143.5	−42.8
13	9	164.5	−41.9	4.5	131	119.5	−55.0
15	9	169.9	−60.7	4.2	152	135.1	−39.7
25	10	306.1	50.5	3.1	250	274.8	25.7
26	10	179.4	−64.8	3.5	187	143.8	−35.8
Mean (excluding #25)	14	171.4	−64.0	5.0	65	137.5 A_{95}=6.1	−35.0 K=43
Mean (for all sites)	15	167.5	−63.7	5.7	46	134.6 A_{95}=7.3	−34.9 K=28

Note: The mean ChRM is given by its declination (D), inclination (I), radius of the 95% confidence cone (α_{95}), and the precision parameter (K) from Fishearian statistics. N indicates the number of samples considered for the mean.

4. Ar–Ar dating

Whole-rock samples from fine-grained mafic dikes of sites #1 (sample 490-02) and #3 (sample 490-03) were dated by ^{40}Ar–^{39}Ar systematics. Samples were irradiated at the IPEN/CNEN IEA-R1 nuclear reactor. Neutron flux gradients were monitored using Fish Canyon sanidine standards. The samples were then stepwise degassed using an Ar-laser system, and the argon isotopic ratios were measured using a MAP-215-50 mass spectrometer at the Centro de Pesquisas Geocronológicas, Universidade de São Paulo; blanks were run between each heating step (see [30] for details on the analytical procedures). Results are given in Table 2.

Apparent ages as a function of cumulative percentage of released ^{39}Ar for both samples are shown in Fig. 8. Sample 490-03 yielded a plateau age of 525±5 Ma calculated from three degassing steps corresponding to 68% of the released gas. The integrated age (529.1± 1.4 Ma) is similar to the plateau age within error (2σ).

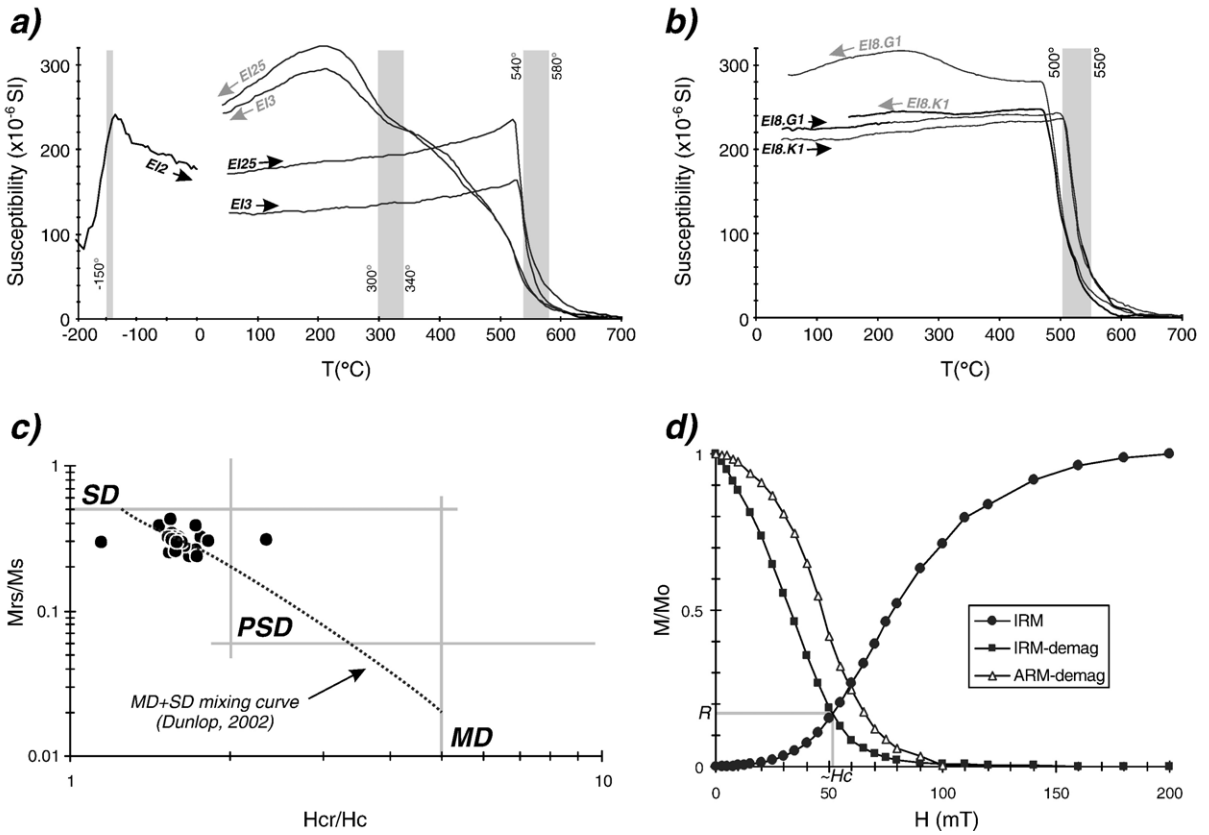


Fig. 6. Magnetic mineralogy of Itabaiana dikes and host rocks. Thermomagnetic curves for (a) dikes and (b) host-rocks. (c) Day plot with indication of SD–MD mixing curve of Dunlop [27]. (d) IRM–ARM acquisition and demagnetization curves.

Sample 490-02 yielded a plateau age of 526±4 Ma calculated from six degassing steps corresponding to 82% of the released gas. These whole-rock Ar–Ar ages correspond to the cooling age of the dikes [31]. Given the fast cooling of basalts and dolerites after dike emplacement in a cool, shallow crust, such ages are here interpreted as a good estimate of the magnetization age of the mafic dikes, which unblocks between 540 and 560 °C for 1 h laboratory delay time during heating (Figs. 3e and 4g).

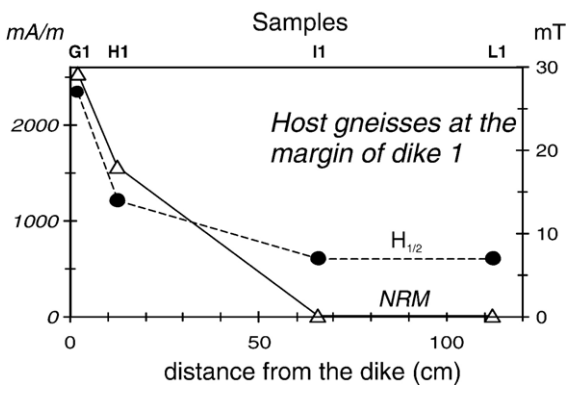


Fig. 7. Variation in NRM and $H_{1/2}$ for the host gneisses when approaching dike 1.

5. Discussion

5.1. A high-quality Early Cambrian paleomagnetic pole for Gondwana

The positive baked contact test and the high stability of the characteristic component, carried by fine single-domain magnetites, indicate that remanence in the Itabaiana dikes is primary, being acquired during the rapid cooling of the intrusive magma. No difference was observed between the two dike sets. Both microphaneritic olivine–basalt and aphanitic basalt dikes (site #7) show similar directions. The moderate dispersion of directions around the mean and the inverse polarity found for one of the dikes (site #25) strongly suggest that the Itabaiana dataset averages out the geomagnetic secular variation.

Table 2
Ar–Ar results

Lab#	Laser (*W*)	$^{40}Ar/^{39}Ar$	$^{38}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$	$^{36}Ar/^{39}Ar$	$^{40}Ar^*/^{39}Ar$	^{40}Ar (mols)	%Rad	Age (Ma)	$\pm 1\sigma$ (Ma)
Sample 0490-02										
0490-02A	0.20	76.80160	0.03674	1.04002	0.02908	68.39692	1.93E-16	88.9	398.2	92.1
0490-02B	0.40	77.74113	0.01590	1.46096	0.02307	71.19449	3.09E-15	91.4	412.8	7.4
0490-02C	0.60	103.50920	0.02083	1.30617	0.02388	96.74603	8.55E-15	93.3	540.5	4.6
0490-02D	0.80	98.69731	0.01326	0.52564	0.01186	95.30919	1.59E-14	96.5	533.6	3.3
0490-02E	1.00	94.48894	0.01378	0.50062	0.00659	92.64964	1.51E-14	98.0	520.7	2.8
0490-02F	1.20	93.80045	0.01231	0.54588	0.00543	92.31364	1.06E-14	98.3	519.0	4.2
0490-02G	1.50	91.42929	0.01028	0.68607	−0.00139	91.99030	1.06E-14	100.5	517.4	3.8
0490-02H	1.80	95.35022	0.00909	1.59397	0.00236	95.00873	8.80E-15	99.4	532.1	4.6
0490-02I	2.21	135.32740	0.01566	5.83556	0.01114	133.68950	6.55E-15	97.9	710.6	7.7
0490-02J	2.70	193.49550	0.04489	8.43458	0.05893	179.06490	3.54E-15	91.3	899.7	21.5
0490-02K	5.01	242.69190	0.01709	13.23766	0.01150	245.33840	1.42E-14	99.0	1144.5	11.0
Sample 0490-03										
0490-03A	0.30	65.22613	0.03316	1.59546	0.05485	49.25965	2.23E-15	75.3	295.4	9.3
0490-03B	0.50	93.77371	0.02554	3.05285	0.03425	84.28268	7.40E-15	89.5	479.3	5.3
0490-03C	0.80	96.87359	0.01435	0.95428	0.00679	95.08047	2.40E-14	98.0	532.5	2.6
0490-03D	1.20	92.32382	0.01131	0.69739	0.00222	91.82049	2.42E-14	99.3	516.6	2.3
0490-03E	1.60	93.56210	0.00994	1.48230	−0.00216	94.52893	1.31E-14	100.8	529.8	3.3
0490-03F	2.20	101.38870	0.00987	3.51446	0.00697	100.14070	1.45E-14	98.2	556.9	3.5
0490-03G	3.50	122.58230	0.01104	6.57629	0.01105	121.05150	6.36E-15	97.7	654.2	8.4
0490-03H	5.01	151.39020	−0.05704	7.35510	−0.03339	163.68390	1.38E-15	106.9	837.8	23.5

The mean remanence direction of the dikes gives a paleomagnetic pole at 134.6° E and 34.9° S (Table 1) for which an age of 525±5 Ma is assigned based on the Ar–Ar results (Table 2). This pole completely satisfies six out of the seven quality criteria proposed by Van der Voo [1] as follows: (1) Ar–Ar ages obtained on two whole-rock samples from different dikes are concordant and likely represent the age of magnetization acquisition; (2) the pole was derived from a set of 136 samples (15 sites) and has adequate statistical precision (A_{95}=7.3 and K=28); (3) remanence vectors were isolated after stepwise AF and thermal demagnetization, both treatments giving similar results; (4) positive baked contact tests performed in three dikes confirm the primary thermoremanent nature of the magnetization; (5) the dikes intruded paleoproterozoic gneisses of the Borborema Province after the peak of deformation and metamorphism, as indicated by regional geochronological data and our own ages; (6) the pole does not match any younger paleomagnetic pole for neither Gondwana nor the independent South American plate after the demise of Pangea. As for the seventh criterion–the presence of reversals–although the record of reversals in the studied dikes can be deduced from the inverse directions found in dike EI25, it alone does not satisfy a reversals test. In sum, the Itabaiana pole may be regarded as a reference pole for the Gondwana supercontinent at Early Cambrian times.

5.2. Implications for the APW path of Gondwana

Several apparent polar wander (APW) paths have been proposed for the Gondwana supercontinent covering the end of Precambrian times and the beginning of the Cambrian. Recently, two slightly different paths were derived by McElhinny et al. [25], Meert [4] and Meert et al. [32], and are represented in Fig. 9. The main difference between them concerns the reference poles chosen to build the curve in the Early to Middle Cambrian times. McElhinny and colleagues relied mostly on Australian data in building their curve, and concede that the Early Cambrian sector is poorly defined in their dataset (see [25], pg. 58). Meert, on the other hand, has tied the same sector of his curve by a different set of poles consisting of the Ntonya Ring pole from southeastern Africa, the poles for São Francisco carbonates of central Brazil, and the Hawker Group poles from southern Australia (C4, C6, C7 and A7; Table 3). But again there are uncertainties on these paleomagnetic data. The very tight grouping of vectors and the absence of reversals suggest that secular variation was probably not averaged out in Ntonya Ring and the corresponding pole would be better interpreted as a virtual geomagnetic pole. Moreover, the ages of poles obtained on São Francisco and Hawker sedimentary successions are not well defined.

The 525±5 Ma Itabaiana pole (C3 in Table 3) provides a strong constraint on the Early Cambrian

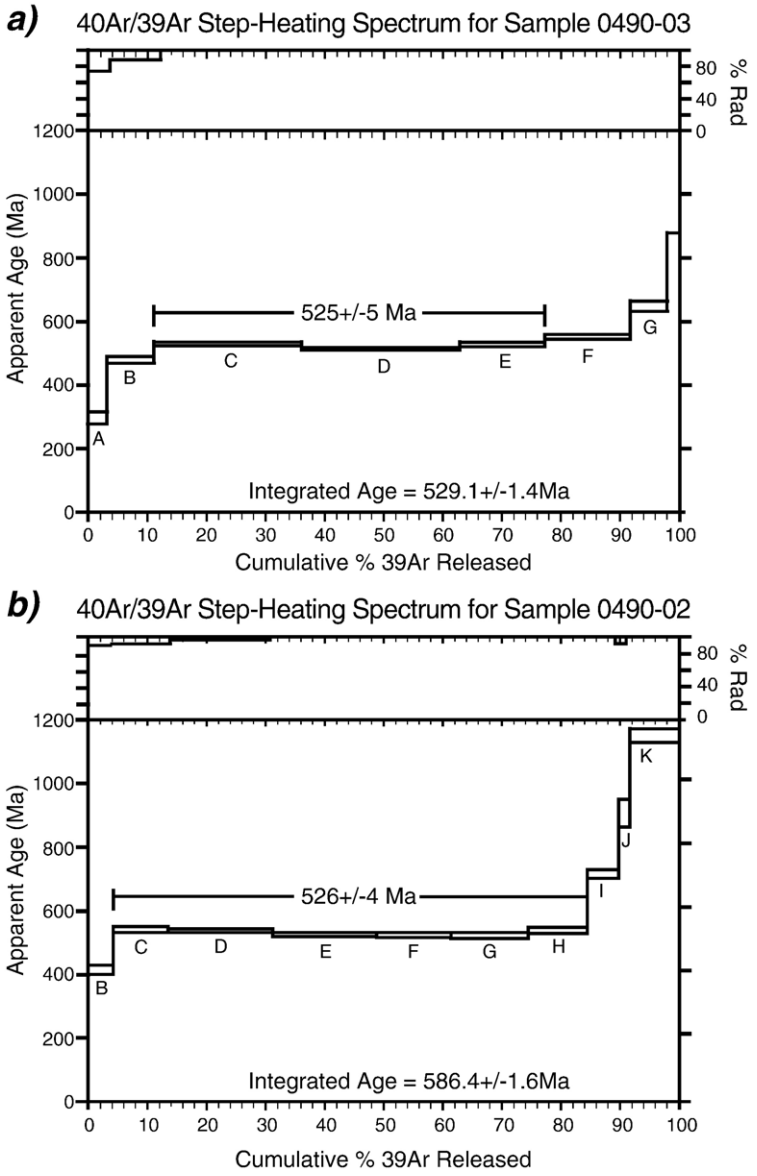


Fig. 8. Apparent ages as a function of cumulative percent ^{39}Ar released for whole-rock samples (a) 490-03 and (b) 490-02. Each step is plotted with 2σ errors. Data are listed in Table 2.

sector of the Gondwana APW path. In Fig. 10a, reference paleomagnetic (south) poles for the 550–500 Ma time-interval rotated to Africa are represented together with their confidence circles. The collection of poles includes poles from Australia, Amazonia, Rio de la Plata, Arabia, Congo (Somalia and Tanzania), São Francisco, Madagascar and Antarctica (see full list in Table 3). The Itabaiana pole falls right to the west of both the Ntonya Ring pole (C4) [39] and the Hawker Group B pole (A7) [34]. The Itabaiana pole also falls close to the poles recently obtained for remagnetized carbonates in the São Francisco (C6) [42], (C7) [43] and Amazon (Am2) [18] cratons for which magnetization ages have been attributed on the basis of whole-rock Pb–Pb data [18,42,43]. All these poles are significantly different from a set of poles dated around 515–500 Ma, which includes the ca. 515 Ma Sor Rondane (AN1) pole [35,54] among others. They also differ from older 550–540 Ma poles, such as those from Sinyai dolerite (C2) [38], Sierra de las Animas (R2) [47], Upper Arumbera sandstone (A5) and Todd River dolomite (A6) [33]. These three clusters of poles define an elbow-shaped apparent polar wander path for the Gondwana supercontinent between 550 and 510 Ma.

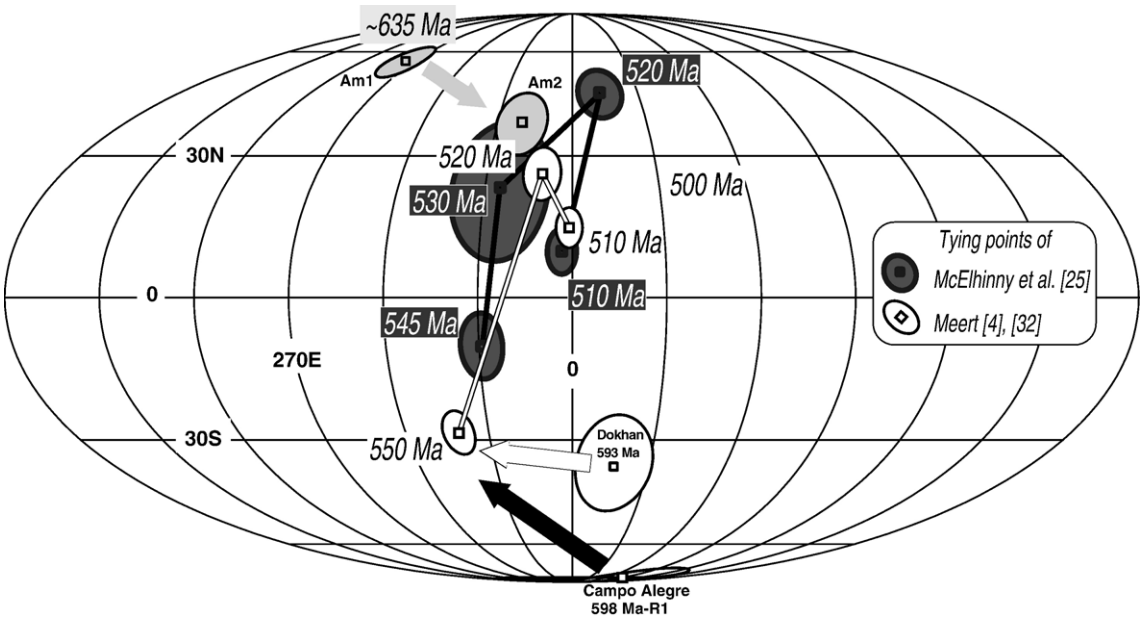


Fig. 9. APW paths for Gondwana between 550–500. Also indicated are poles for the Amazon craton (Am1 and Am2).

For the sake of comparison, poles of Table 3 were plotted in Fig. 10b and c after different Gondwana configurations. In Fig. 10b we have used the Lowver and Scotese [52] reconstruction. In Fig. 10c we have used the 'tight fit' of Reeves et al. [53]. The latter takes into account pre-Mesozoic deformation within the African plate. The two reconstructions produce slightly different pole distributions; poles fitting better an APW curve in the 'tight fit' configuration. It is worth noting that, in spite of the different pinning-points and the different reconstructions used, all paths (including our own path) imply fast drift rates for Gondwana in the Early Cambrian interval, varying from 24 to 31 cm/yr.

5.3. Implications for Gondwana assembly

The paths indicated in Figs. 9 and 10 reveal a coincidence of poles from cratonic units and encircling fold belts after 550 Ma into a united West Gondwana configuration. The similar path found for these units corroborates collisional ages older than 550 Ma found between these blocks (see [5] and references therein). The actual timing of these collisions could be unveiled by the comparison of older paleomagnetic poles for these and other cratons. However, only three poles are available for units with ages between 700 and 550 Ma (Table 3). The Dokhan volcanics pole (C1) [36] is the most reliable record among them. The age of this pole is constrained for 593±15 Ma after U–Pb dating [55]. However, this pole was obtained in Egypt which is not part of the Congo Craton and might have had an independent kinematic evolution before joining Gondwana. In the West Africa craton a pole was obtained for the Adma intrusion (W1) dated by two U–Pb ages at 616 ±11 and 613±3 Ma [48]. For a similar age, a pole on the volcanic rocks from Campo Alegre (R1) [46] dated at 598±29 Ma [56] constrains the position of the Rio de la Plata craton. In contrast to the younger Cambrian poles, these data are significantly different for overlapping ages, and point to a complex accretion of West Gondwana prior to 550 Ma ago. In addition, a very stable two-polarity component identified in the Araras Group basal dolomites of Amazonia (pole Am1, [18,57]) departs strongly from the other poles. The age of this pole is still loosely constrained on the basis of Sr and C isotopes correlation to 630–600 Ma post-Marinoan cap carbonates [58,59]. This age is further constrained by a 625 Ma K–Ar age obtained on mafic sills that cut across correlative units at the Tucavaca belt in Bolivia at the southern margin of the Amazon Craton [60]. A younger pole (Am2) obtained in the same study fails a fold test. It matches the Early Cambrian poles of remagnetized carbonates from the São Francisco craton, as well as poles from Ntonya Ring, Hawker Group and Itabaiana.

Taking into account the current paleomagnetic database as well as geological and geochronological data, we have tentatively represented the assembly of major Gondwana pieces in Fig. 11 by two paleogeographic

Table 3
Selected paleomagnetic poles for Gondwana and Laurentia between 600 and 500 Ma

CLPOL code		Plat (°N)	Plon. (°E)	Regular fit		Tight fit		α_{95} (°)	Age (Ma)	Reference no.
				Plat. (°N)	Plon. (°E)	Plat. (° N)	Plon. (°E)			
Australia (AU)										
A1	Upper Arumbera	−46	337	−9	332	−14	336	5	545	[33]
A2	Todd River	−43	340	−5	331	−10	335	8	528	[33]
A3	Hawker Group	−21	15	28	348	24	348	13	535	[34]
A4	Pertaoorta Group	−33	12	15	349	12	351	8	520	[34]
A5	Kangaroo Island	−34	15	15	352	12	354	12	519	[34]
A6	Billy Creek	−37	20	13	357	11	358	14	517	[34]
A7	Giles Creek	−38	25	13	1	11	3	10	507	[34]
A8	Lower Lake Frome	−31	27	20	2	18	3	10	507	[34]
East Antarctica (EA)										
AN1	Sor Rondane	−29	10	11	9	10	7	5	510	[35]
Africa and central South America										
C1	Dokhan volcanics (SO)	43	216	43	216	42	227	5	593	[36]
C2	Mirbat SS (AR)	−32	334	−32	334	−34	330	2	550	[37]
C3	**Sinyai dolerite (SO)**	−28	319	−28	319	−40	321	5	547	[38]
C4	**Itabaiana dikes (SA)**	35	315	31	331	29	330	8	525	*
C5	Ntonya Ring (TA)	28	345	28	345	28	345	2	522	[39,40]
C6	Madagascar virgation zone (MA)	−7	353	12	352	13	350	14	521	[41]
C7	Bambui+ Salitre, C (SA)	32	322	34	338	32	337	3	525	[42]
C8	Bambui B (SA)	15	331	27	359	26	357	3	520	[43]
C9	Juiz de Fora Complex (SA)	4	320	12	358	10	357	10	510	[44]
C10	Carion Granite (MA)	−7	1	13	360	14	358	11	508	[32]
C11	Piquete Formation (SA)	−1	347	24	24	24	22	10	500	[45]
W1	**Adma Diorites**	34	344	34	344	31	341	9	615	[48]
Rio de la Plata (RP)										
R1	Campo Alegre	−57	223	−82	52	−81	67	9	590	[46]
R2	Sierra de las Animas 2	−17	251	−47	312	−50	312	16	550	[47]
R3	Sierra de las Animas 1	6	338	25	11	24	9	19	520	[47]
Amazonia (SA)										
Am1	Araras Group A	83	113	51	249	49	251	9	600	[18]
Am2	Araras Group B	34	327	38	340	36	338	10	525	[18]
Laurentia (LA)										
L1	**Skinner Cove Fm**	−15	337			550[a]		9	550[a]	[49]
L2	**Tapeats Sandstones**	−5	338			525[a]		3	525[a]	[50]

Notes:
*This work.
[a]Ages according to [49] and [51].
Regular fit (Euler rotation poles to Africa as in [52]). Australia (AU): 24.5°N, 112.3°E, −56.3°, East Antarctica (EA): −7.78°N, −31.42°E, 58°, South America (SA): 45.5 °N, −32.2°E, 58.2°, Madagascar (MA): −3.41°N, −81.7°E, 19.73°.
Tight fit (Euler rotation poles as in [53]): Arabia (AR): −59.43°N, 85.67°E, 4.107°; Somalia (SO): 1.59°N, 36.67°E, 11.99°; Tanzania (TA): −89.0°N, 210.0°E, 0.4°; Madagascar (MA): −6.5°N, 278.73°E, 21.193°; East Antarctica (EA): −10.36°N, 329.75°E, 59.999°, South America (SA): 43.017°N, 329.935°E, 58.842°; Australia (AU): −25.505°N, 299.332°E, 56.262°; West Africa (WA): 8°N, 9°E, 7°. Poles AS, AU and WA were given by Colin Reeves (unpublished).
Paleomagnetic poles in bold were used in reconstructions of Fig. 10.

reconstructions at 550 and 525 Ma. In the reconstructions, major orogenic belts are represented by gray swathes and include the Mozambique (1), Brasiliano–Panafrican (1′), Kuunga (2) and Pampean–Araguaia (3) belts. They separate several continental pieces merged together along diachronic events spanning in

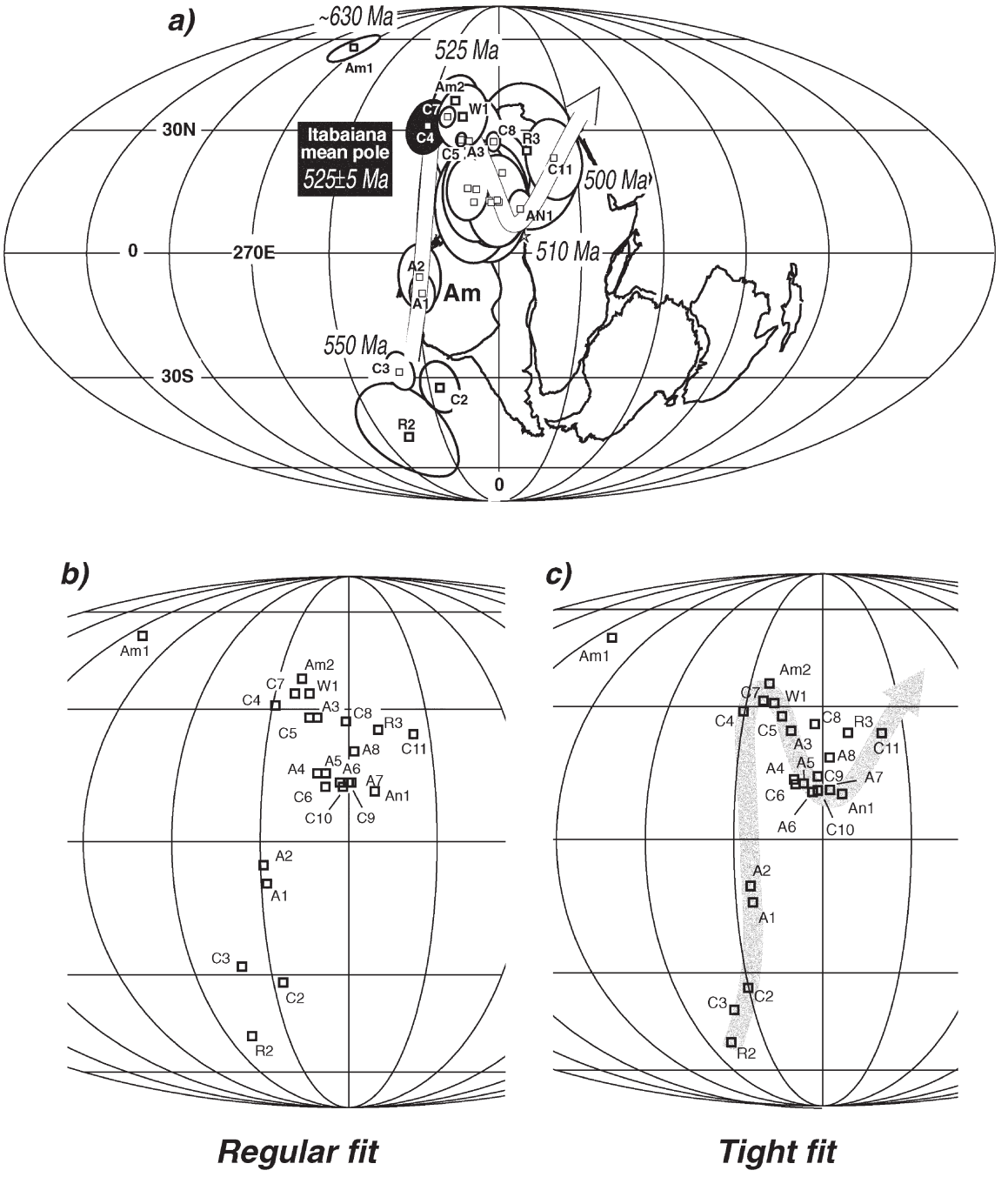


Fig. 10. (a) APW path for Gondwana incorporating the Itabaiana pole. Below, the 'regular fit' (b) of Lowver and Scotese [52] and the 'tight fit' (c) of Reeves et al. [53] are shown. Note the better grouping of poles obtained in the latter fit, which takes into account pre-Mesozoic deformation within the African plate.

time the whole Neoproterozoic and also the Early Cambrian.

In the first reconstruction (Fig. 11a) the paleogeography for 550 Ma is represented by a huge continental mass (hereafter referred as proto-Gondwana) to the west. The proto-Gondwana includes the continental fragments around Congo–São Francisco and the Australia–Antarctica–Kalahari block. These cratons have merged through the Kuunga orogenic belts [4] which encompasses the collisional belts of Damara–Gariep in southern Africa, the south and central Madagascar, Sri Lanka, the Maud Province, Lützow–

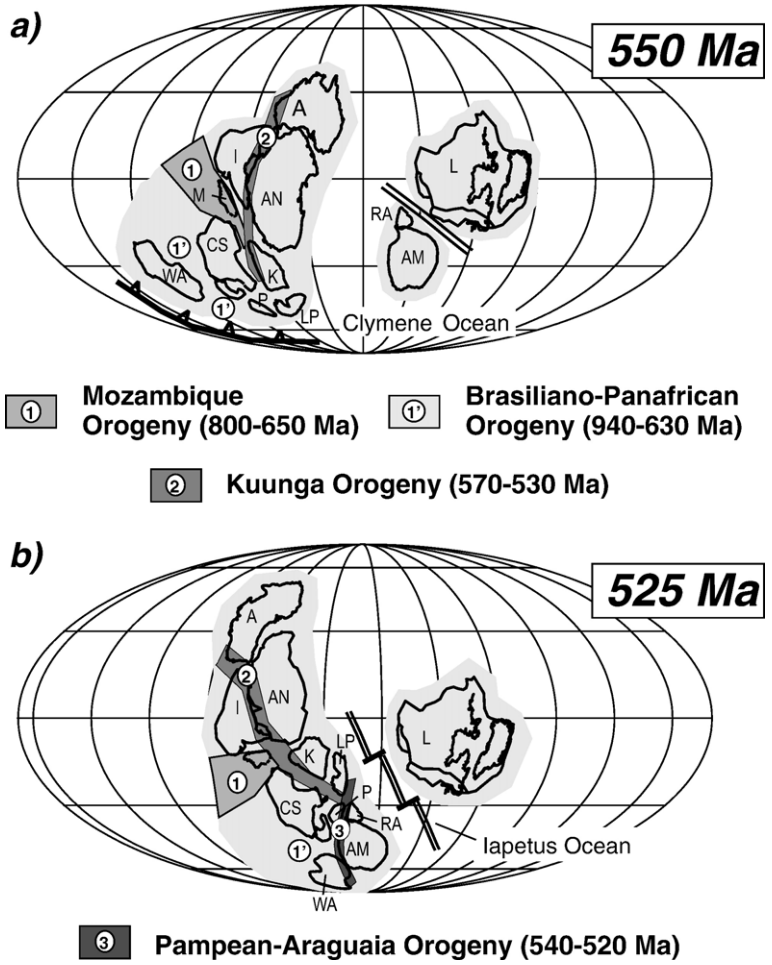


Fig. 11. Paleogeographic reconstructions for (a) 550 Ma, and (b) 525 Ma ago. WA: West Africa, CS: Congo-São Francisco, P: Parana, K: Kalahari, LP: Rio de la Plata, AN: Antarctica, A: Australia, M: Madagascar, I: India, AM: Amazonia, RA: Rio Apa, L: Laurentia. The ocean between Amazonia, Rio Apa, Pampia and the proto-Gondwana was baptized Clymene in here after the wife of the titan Iapetus in Greek mythology.

Holm and Prydz Bay in East Antarctica, and the Darling belt of western Australia. The position of this block is constrained by the Sinyai dolerite pole dated at 547± 4 Ma [38]. To the east, Amazonia is represented close to Laurentia. Previous works have constrained the rifting between these cratons by comparison between the Laurentia and Gondwana paleomagnetic data (e.g., [57]). However, the Puga poles recently obtained for Amazonia [18] indicate that this unit has not joined Gondwana until Cambrian times. Therefore, we suggest that separation between Amazonia and Laurentia is more likely to have started at 550 Ma, when plume-related volcanic activity in the eastern Laurentian margin peaked [61] coincidentally with the transition from rift to drift-related sedimentation in coastal basins [62,63]. The position of Laurentia and Amazonia at the putative beginning of the rifting is constrained by the Skinner Cove pole [49,51] dated at $550.5^{+3}/_{-2}$ Ma (U–Pb zircon; [61]).

After separation from Laurentia, Amazonia and other minor adjoining blocks would collide with the proto-Gondwana at Cambrian times forming a major orogenic belt, comprising the Araguaia, Paraguay and Pampean belts. Contrary to previous models in which such Cambrian events are local in extent, we advocate that they represent indeed a major orogenesis shaping the West Gondwana. The collisional age along the Araguaia–Pampean belt is constrained by a series of 550–500 Ma U–Pb dates obtained on syn-tectonic granites from Araguaia [64] to western Argentina [16], as well as recent 540–530 Ma Ar–Ar cooling ages in biotites obtained on metavolcanics in the Paraguay belt [65]. Docking of several smaller terrains that formed the western margin of Gondwana followed this

major collision (e.g., [66–68]). The paleogeographic reconstruction of Gondwana just after its final suturing is now well constrained by the Itabaiana pole.

Acknowledgments

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References

[1] R. Van der Voo, The reliability of paleomagnetic data, Tectonophysics 184 (1990) 1–9.
[2] M.O. McWilliams, Palaeomagnetism and Precambrian tectonic evolution of Gondwana, in: A. Kröner (Ed.), Precambrian Plate Tectonics, vol. 26, Elsevier, Amsterdam, 1981, pp. 649–687, Chapter.
[3] R.J. Stern, Crustal evolution in the East African orogeny: a neodymium isotopic perspective, J. Afr. Earth Sci. 34 (2002) 109–117.
[4] J.G. Meert, A synopsis of events related to the assembly of eastern Gondwana, Tectonophysics 362 (2003) 1–40.
[5] U.G. Cordani, K. Sato, W. Teixeira, C.C.G. Tassinari, M.A.S. Basei, Crustal evolution of the South American platform, in: U. G. Cordani, E.J. Milani, A. Thomaz Filho, D.A. Campos (Eds.), Tectonic Evolution of South America, 31st International Geological Congress, Rio de Janeiro, Brazil, 2000, pp. 19–40.
[6] R. Trompette, Neoproterozoic (~600 Ma) aggregation of Western Gondwana: a tentative scenario, Precambrian Res. 82 (1997) 101–112.
[7] B.B. Brito Neves de, M.C. Campos Neto, R.A. Fuck, From Rodinia to Western Gondwana: an approach to the Brasiliano–Pan African cycle and orogenic collage, Episodes 22 (3) (1999) 155–166.
[8] M.M. Pimentel, R.A. Fuck, N.F. Botelho, Granites and the geodynamic history of the neoproterozoic Brasília belt, Central Brazil: a review, Lithos 46 (1999) 463–483.
[9] U.G. Cordani, M.S. D'Agrella-Filho, B.B. Brito-Neves, R.I.F. Trindade, Tearing up Rodinia: the Neoproterozoic palaeogeography of South American cratonic fragments, Terra Nova 15 (2003) 350–359.
[10] R.S. Schmitt, R.A.J. Trouw, W.R. Van Schmus, M.M. Pimentel, Late amalgamation in the central part of West Gondwana: new geochronological data and the characterization of a Cambrian collisional orogeny in the Ribeira Belt (SE Brazil), Precambrian Res. 133 (2004) 29–61.
[11] C.M. Valeriano, N. Machado, A. Simonetti, C.S. Valladares, H.J. Seer, L.S.A. Simões, U–Pb geochronology of the southern Brasília belt (SE Brazil): sedimentary provenance, Neoproterozoic orogeny and assembly of West Gondwana, Precambrian Res. 130 (2004) 27–55.
[12] J.H. Laux, M.M. Pimentel, E.L. Dantas, R. Armstrong, S.L. Junges, Two neoproterozoic crustal accretion events in the Brasília belt, central Brazil, J. South Am. Earth Sci. 18 (2005) 183–198.
[13] R. Caby, Precambrian terranes of Benin–Nigeria and northeast Brazil and the Late Proterozoic south Atlantic fit, Spec. Pap. - Geol. Soc. Am. 230 (1989) 145–158.
[14] W.R. Van Schmus, B.B. Brito Neves, P.C. Hackspacher, M. Babinski, U/Pb and Sm/Nd geochronologic studies of eastern Borborema Province, northeastern Brazil: initial conclusions, J. South Am. Earth Sci. 8 (1995) 267–288.
[15] C.J. Archanjo, R.I.F. Trindade, J.L. Bouchez, M. Ernesto, Granite fabrics and regional-scale strain partitioning in the Seridó belt (Borborema province, NE Brazil), Tectonics 21 (2002), doi:10.1029/2000TC001269.
[16] C.W. Rapela, R.J. Pankhurst, C. Casquet, E. Baldo, J. Saavedra, C. Galindo, Early evolution of the proto-Andean margin of South America, Geology 26 (1998) 707–710.
[17] E. Tohver, M.S. D'Agrella-Filho, R.I.F. Trindade, Paleomagnetic record of Africa and South America for the 1200–500 Ma interval, and evaluation of Rodinia and Gondwana assemblies, Precambrian Research (in press).
[18] R.I.F. Trindade, E. Font, M.S. D'Agrella-Filho, A.C.R. Nogueira, C. Riccomini, Low-latitude and multiple geomagnetic reversals in the Neoproterozoic Puga cap carbonate, Amazon craton, Terra Nova 15 (2003) 441–446.
[19] W.R. Van Schmus, B.B. Brito-Neves, I.S. Williams, P.C. Hackspacher, A.H. Fetter, E.L. Dantas, M. Babinski, The Seridó Group of NE Brazil, a late Neoproterozoic pre- to syn-collisional basin in West Gondwana: insights from SHRIMP U–Pb detrital zircon ages and Sm–Nd crustal residence (TDM) ages, Precambrian Res. 127 (2003) 287–327.
[20] M. Corsini, L. Lambert de Figueiredo, R. Caby, G. Feraud, G. Ruffet, A. Vauchez, Thermal history of the Pan-African/Brasiliano Borborema Province of Northeast Brazil deduced from $^{40}Ar/^{39}Ar$ analysis, Tectonophysics 285 (1998) 103–117.
[21] B.B. Brito Neves, W.R. Van Schmus, A. Fetter, Noroeste da África–Nordeste do Brasil (Província Borborema). Ensaio comparativo e problemas de correlação, Revista do Instituto de Geociências – USP – Série Científica 1 (2001) 59–78.
[22] E.J. Santos, A.P. Nutman, B.B. Brito Neves, Idades SHRIMP U–Pb do Complexo Sertânia: implicações sobre a evolução tectônica da Zona Transversal, Província Borborema, Revista do Instituto de Geociências Geol. USP Sér. Cient., São Paulo 4 (2004) 1–12.
[23] J.L. Kirschvink, The least-squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc. 62 (1980) 699–718.
[24] P. McFadden, M.W. McElhinny, Classification of the reversal test in paleomagnetism, Geophys. J. Int. 103 (1990) 725–729.
[25] M.W. McElhinny, C.McA. Powell, S.A. Pisarevsky, Paleozoic terranes of eastern Australia and the drift history of Gondwana, Tectonophysics 362 (2003) 41–65.
[26] M.I.B. Raposo, M. Ernesto, P.R. Renne, Paleomagnetism and $^{40}Ar/^{39}Ar$ dating of the Early Cretaceous Florianópolis dike swarm (Santa Catarina Island), Southern Brazil, Phys. Earth Planet. Inter. 108 (1998) 275–290.
[27] D.J. Dunlop, Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc) 2. Application to data for rocks, sediments, and soils, J. Geophys. Res. 107 (2002) B3, doi:10.1029/2001JB000487.
[28] S. Cisowski, Interacting vs. non-interacting single domain behavior in natural and synthetic samples, Phys. Earth Planet. Inter. 26 (1981) 56–62.
[29] D.J. Dunlop, Ö. Özdemir, Effect of grain size and domain state on thermal demagnetization, Geophys. Res. Lett. 27 (2000) 1311–1314.
[30] P.M. Vasconcelos, A.T. Onoe, K. Kawashita, A.J. Soares, W. Teixeira, $^{40}Ar–^{39}Ar$ geochronology at the Instituto de Geociências, USP: instrumentation, analytical procedures, and

calibration, An. Acad. Bras. Ciênc. ISSN :0001-3765 74 (2002) 297–342.
[31] I. McDougall, T.M. Harrisson, Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ Method, Oxford University Press, Oxford, 1999, 269 pp.
[32] J.G. Meert, A. Nédélec, C. Hall, M.T.D. Wingate, M. Rakotondrazafy, Paleomagnetism, geochronology and tectonic implications of the Cambrian-age Carion granite, Central Madagascar, Tectonophysics 340 (2001) 1–21.
[33] J.L. Kirschvink, The Precambrian–Cambrian boundary problem: paleomagnetic directions from the Amadeus basin, Central Australia, Earth Planet. Sci. Lett. 40 (1978) 91–100.
[34] C.T. Klootwijk, Early Paleozoic paleomagnetism in Australia, Tectonophysics 64 (1980) 249–332.
[35] J.D.A. Zijderveld, Natural remanent magnetizations of some instrusive rocks from the Sor Rondane mountains, Queen Maud Land, Antarctica, J. Geophys. Res. 73 (1968) 3773–3785.
[36] J. Davies, A.E.M. Nairn, R. Ressetar, The paleomagnetism of certain Late Precambrian and Early Paleozoic rocks from the Red Sea Hills, Eastern Desert, Egypt, J. Geophys. Res. 85 (1980) 3699–3710.
[37] O. Kempf, P. Kellerhals, W. Lowrie, A. Matter, Paleomagnetic directions in Late Precambrian glaciomarine sediments of the Mirbat Sandstone Formation, Oman, Earth Planet. Sci. Lett. 175 (2000) 181–190.
[38] J.G. Meert, R. Van der Voo, Paleomagnetic and $^{40}Ar/^{39}Ar$ study of the Sinyai dolerite, Kenya: implications for Gondwana assembly, J. Geol. 104 (1996) 131–142.
[39] J.C. Briden, Paleomagnetism of the Ntonya Ring Structure, Malawi, J. Geophys. Res. 73 (1968) 725–733.
[40] J.C. Briden, E. McClelland, D.C. Rex, Proving the age of a paleomagnetic pole: the case of the Ntonya Ring structure, Malawi, J. Geophys. Res. 98 (1993) 1743–1749.
[41] J.G. Meert, A. Nédélec, C. Hall, The stratoid granites of central Madagascar: paleomagnetism and further age constraints on Neoproterozoic deformation, Precambrian Res. 120 (2003) 101–129.
[42] R.I.F. Trindade, M.S. D'Agrella-Filho, M. Babinski, E. Font, B. B. Brito-Neves, Paleomagnetism and geochronology of the Bebedouro cap carbonate: evidence for continental-scale Cambrian remagnetization in the São Francisco craton, Brazil, Precambrian Res. 128 (2004) 83–103.
[43] M.S. D'Agrella-Filho, M. Babinski, R.I.F. Trindade, W.R. Van Schmus, M. Ernesto, Simultaneous remagnetization and U–Pb isotope resetting in Neoproterozoic carbonates of the São Francisco craton, Brazil, Precambrian Res. 99 (2000) 179–196.
[44] M.S. D'Agrella-Filho, M.I.B. Raposo, M. Egydio-Silva, Paleomagnetic Study of the Juiz de Fora Complex, SE Brazil: implications for Gondwana, Gondwana Res. 7 (2004) 103–113.
[45] M.S. D'Agrella-Filho, I.G. Pacca, K. Sato, Paleomagnetism of metamorphic rocks from the Piquete Region—Ribeira Valley, Southeastern Brazil, Rev. Bras. Geofís. 4 (1986) 79–84.
[46] M.S. D'Agrella-Filho, I.G. Pacca, Paleomagnetism of the Itajaí, Castro and Bom Jardim Groups from Southern Brazil, Geophys. J. 93 (1988) 365–376.
[47] L. Sánchez-Bettucci, A.E. Rapalini, Paleomagnetism of the Sierra de Las Animas Complex, southern Uruguay: its implications in the assembly of western Gondwana, Precambrian Res. 118 (2002) 243–265.
[48] P. Morel, Palaeomagnetism of a Pan-African diorite: a Late Precambrian pole for western Africa, Geophys. J. R. Astron. Soc. 65 (1981) 493–503.
[49] P.J.A. McCausland, J.P. Hodych, Paleomagnetism of the 550 Ma Skinner Cove volcanics of western Newfoundland and the opening of the Iapetus Ocean, Earth Planet. Sci. Lett. 163 (1998) 15–29.
[50] D.P. Elston, S.L. Bressler, Paleomagnetic poles and polarity zonation from Cambrian and Devonian strata of Arizona, Earth Planet. Sci. Lett. 36 (1977) 423–433.
[51] J.P. Hodych, R.A. Cox, J. Kosler, An equatorial Laurentia at 550 Ma confirmed by Grenvillian inherited zircons dated by LAM ICP-MS in the Skinner Cove volcanics of western Newfoundland: implications for inertial interchange true polar wander, Precambrian Res. 129 (2004) 93–113.
[52] L. Lowver, C.R. Scotese, A revised reconstruction of Gondwanaland, in: G.D. McKenzie (Ed.), Gondwana Six: Structure, Tectonics and Geophysics, Am. Geophys. Un. Mon., vol. 40, 1987, pp. 17–23.
[53] C.V. Reeves, M.J. de Wit, B.K. Sahu, Tight reassembly of Gondwana exposes Phanerozoic shears in Africa as global tectonic players, Gondwana Res. 7 (2004) 7–19.
[54] A.M. Grunow, Implications for Gondwana of new Ordovician paleomagnetic data from igneous rocks in southern Victoria Land, East Antarctica, J. Geophys. Res. 100 (1995) 12589–12603.
[55] S.A. Wilde, K. Youssef, Significance of SHRIMP U–Pb dating of the Imperial Porphyry and associated Dokhan volcanics, Gebel Dokhan, north Eastern Desert, Egypt, J. Afr. Earth Sci. 31 (2000) 403–413.
[56] M.A.S. Basei, S.B. Citroni, O. Siga Jr., Stratigraphy and age of Fini-Proterozoic Basins of Paraná and Santa Catarina States, Southern Brazil, Bol. Inst. Geociênc. 29 (1998) 195–216.
[57] E. Font, R.I.F. Trindade, A. Nédélec, Detrital remanence magnetization in hematite-bearing Neoproterozoic Puga cap dolostone, Amazon Craton: a rock magnetic and SEM study, Geophys. J. Int. 163 (2005) 491–500.
[58] K.H. Hoffmann, D.J. Condon, S.A. Bowring, J.L. Crowley, U–Pb zircon date from the Neoproterozoic Ghaub Formation, Namibia: constraints on Marinoan glaciation, Geology 32 (2004) 817–820, doi:10.1130/G20519.1.
[59] D. Condon, M. Zhu, W. Bowring, A. Yang, Y. Jin, U–Pb Ages from the Neoproterozoic Doushantuo Formation, China, Science 308 (2005) 95–98.
[60] M. Litherland, R.N. Annels, D.P.F. Derbyshire, C.J.N. Fletcher, M.P. Hawkins, B.A. Klink, W.I. Mitchell, E.A. O´Connor, P.E.J. Pitfield, G. Power, B.C. Webb, The Proterozoic of eastern Bolivia and its relationship to the Andean mobile belt, Precambrian Res. 43 (1989) 157–174.
[61] P.A. Cawood, P.J.A. McCausland, G.R. Dunning, Opening Iapetus: constraints from the Laurentian margin in Newfoundland, GSA Bull. 113 (2001) 443–453.
[62] J.H. Puffer, A late Neoproterozoic eastern Laurentian superplume: location, size, chemical composition, and environmental impact, Am. J. Sci. 302 (2002) 1–27.
[63] H. Williams, R.N. Hiscott, Definition of the Iapetus rift–drift succession in western Newfoundland, Geology 15 (1987) 1044–1047.
[64] C.A.V. Moura, H.E. Gadette, Evidence of Brasiliano/Pan-African deformation in the Araguaia belt: implication for Gondwana evolution, Rev. Bras. Geociênc. 23 (1993) 117–123.
[65] M.C. Geraldes, C.C.G. Tassinari, A.T. Onoe, $^{40}Ar–^{39}Ar$ metamorphic record of a collision related to the Western Gondwana collage: the (541–531 Ma) Paraguay Belt in the Nova Xavantina (MT) region, Simpósio Nacional de Estudos Tectônicos, Buzios, Anais do IX SNET, 2003, pp. 54–57.

[66] W.A. Thomas, R.A. Astini, P.A. Mueller, G.E. Gehrels, J.L. Wooden, Transfer of the Argentine Precordillera terrane from Laurentia: constraints from detrital-zircon geochronology, Geology 32 (2004) 965–968.
[67] A.E. Rapalini, C.A. Cingolani, First late Ordovician paleomagnetic pole for the Cuyania (Precordillera) Terrane of Western Argentina: a microcontinent or a Laurentian Plateau? Gondwana Res. 7 (2004) 1089–1104.
[68] J.B. Murphy, S.E. Pisarevsky, R.D. Nance, J.D. Keppie, Neoproterozoic–Early Paleozoic evolution of peri-Gondwanan terranes: implications for Laurentia–Gondwana connections, Int. J. Earth Sci. (Geol Rundsch) 93 (2004) 659–682.